

SAMPREETH AVVARI

New York, NY | (908) 307-7763 | spa9659@nyu.edu | LinkedIn | GitHub | Portfolio | Blog

SUMMARY

AI Engineer who ships LLM and ML products to production end to end, from prompt and agent design through evaluation and serving. Built 5 production systems solo, replacing a \$98K vendor quote and cutting model hallucinations 35% under HIPAA-eligible controls.

EDUCATION

MS, Computer Engineering, New York University, NY (CGPA 3.84/4)

Aug 2023 - May 2025

BTech, Computer Science and Engineering, JNTU, India (CGPA 3.92/4)

Jul 2017 - Jun 2021

SKILLS

AI / LLM: LLM product development, GenAI, prompt engineering, context engineering, JSON Schema contracts, agentic workflows (tool use, multi-step orchestration), evals, rubrics, guardrails, RAG (pgvector, Milvus), embeddings, fine-tuning, RLHF (GRPO/PPO), QLoRA, PyTorch, Transformers, Vertex AI Gemini, Llama, LangChain

Languages & Full-stack: Python, TypeScript, JavaScript, Go, Java, SQL, C++; FastAPI, Flask, Node.js, REST, GraphQL, microservices, React, Next.js

Data & Cloud / MLOps: PostgreSQL, pgvector, MongoDB, Redis, BigQuery, Kafka; GCP (Cloud Run, Pub/Sub), AWS (S3, RDS, MSK), Docker, Kubernetes, Terraform, MLflow, Airflow, CI/CD, monitoring (Prometheus, Grafana, OpenTelemetry)

Applied-AI delivery: LLM products from prototype to production, agent design, eval-driven development, guardrails and safety, model selection, fast iteration with stakeholders; Google Gen AI Leader

WORK EXPERIENCE

AI Engineer | Hybridge Implants, New York

Sep 2025 - Present

- Partnered with clinicians and the C-suite to ship 5 production ML and LLM systems solo, taking models from prototype to production and replacing a \$98K vendor quote.

Doc Coach, LLM consultation-QA pipeline

- Shipped Doc Coach, an LLM consultation-QA pipeline (Vertex AI Gemini) scoring every consult on a 7-criterion rubric, with versioned prompts and JSON Schema guardrails cutting hallucinations 35%.
- Raised treatment acceptance 130% and downstream revenue 43% through eval-driven prompt iteration under HIPAA-eligible controls.

CBCT Scan Validator, in-house dental-CT quality classifier

- Built an in-house CBCT quality classifier (frozen nnU-Net v2 encoder, OpenVINO, 5.5s/scan) flagging the 10-15% of 600 monthly scans unusable for implant design.
- Gated releases on a 20-scan holdout that exposed a leaky 0.80 AUROC, shipping an honest 0.63 with a human-review queue over an inflated number.

Treatment Estimator, Next.js and Postgres quoting platform

- Delivered a Next.js and Postgres quoting platform (5-model pricing engine, coordinator wizard) replacing a decade-old vendor tool, now serving 50 consults a week.
- Hardened it with Postgres-enforced price locks (write-once invariants, triggers) and 100% branch tests, sustaining daily use under \$35 a month.

Software Engineer | Shure Incorporated, India

Aug 2021 - Aug 2023

- Engineered event-driven Flask REST APIs on AWS ingesting 500K+ daily Kafka events, cutting processing time 20% with async workers.
- Directed delivery across 3 release tracks with Jenkins canary deploys, cutting deploy errors 40% with 150+ pytest and Selenium tests at 90%+ coverage, and led code reviews.

RESEARCH EXPERIENCE

Machine Learning Researcher | New York University (writeup)

May 2024 - Sep 2025

- Fine-tuned Llama 3.1 8B through a full RLHF lifecycle (SFT then GRPO/PPO, QLoRA reward models), reaching a 67% win rate on human evaluation.
- Engineered a Python and Airflow ETL turning 118 Reddit CMV shards into 38K preference pairs with schema enforcement and failure alerting, tracked in WandB.

PROJECTS

JobPilot, open-source (MIT) AI job-hunt autopilot

- Built JobPilot, an open-source job-hunt autopilot adopted by 12 peers, AI-scoring fit and auto-tailoring an ATS-clean resume per match across 11 sources for under \$10 a month.
- Designed 7 agentic patterns: an LLM-as-judge scorer, JSON-schema output contracts, a self-refining rewrite loop, a RAG assistant, multi-step orchestration, tool-use across Gmail, Sheets, and Drive, and guardrails that never auto-send.

Loan Radar, self-retraining loan-default scoring platform

- Architected Loan Radar, a self-retraining loan-default platform on Terraform Kubernetes (Ray) with 3 FastAPI microservices, Docker, and MLflow-tracked CI/CD for rollback-safe predictions.